

Hafiz Abdullah Qasim

 Johar Town, Lahore (PK)  +92 309 9771137  abdullahqasim607@gmail.com  [abdullah-qasim-dev](https://www.linkedin.com/in/abdullah-qasim-dev)  [abdullah-qasim-dev](https://github.com/abdullah-qasim-dev)

Professional Summary:

Data Science student at FAST NUCES with strong foundation in Python, C++, and machine learning. Experienced in building predictive models, data pipelines, and desktop applications, with hands-on work in stock prediction, customer churn analysis, and NLP-based systems. Skilled in SQL, data structures & algorithms, and problem solving, with a growing focus on AI-driven solutions.

Education:

❖ Bachelors in Data Science FAST NUCES, Lahore	08/2023 -2027 (Current)
❖ ICS Punjab College, Lahore	(2021 – 2023)
❖ Matric The Trust School, Lahore	(2019 – 2021)

Skills:

- ❖ Languages: Python, C++, C#, SQL
- ❖ Programming Concepts: OOP, Data Structures & Algorithms, Complexity Analysis
- ❖ ML & Data Science: scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Regression, Classification, Clustering, Model Evaluation
- ❖ NLP & LLMs: Hugging Face, Prompt Engineering, Text Processing, BLEU/ROUGE Evaluation
- ❖ Data Engineering: PostgreSQL, ETL/ELT Pipelines, Dimensional Modeling, Data Warehousing

Projects:

- ❖ **LLM-Based Personalized K-12 Tutor:** Built an NLP system using Flan-T5 to generate personalized feedback for K-12 short-answer questions. Evaluated different prompt strategies using BLEU/ROUGE scores and human assessment, with analysis of model errors and student misconceptions.
- ❖ **Stock Market Price Prediction Model:** Built a machine-learning model to forecast Apple Inc. stock prices using historical data. Applied Linear, Polynomial, Lasso Regression, and Random Forest, extracting technical features and evaluating performance with MSE, MAE, and R² metrics.
- ❖ **Stock & Crypto Data Warehouse:** Implemented ETL and ELT pipelines in PostgreSQL with dimensional modeling and evaluated performance and cost implications of different SQL join strategies.
- ❖ **Customer Churn Prediction System:** Built a machine-learning model to predict customer churn using telecom data. Performed data preprocessing, feature encoding, and trained classification models to identify customers likely to leave the service. Evaluated models using standard performance metrics.
- ❖ **Clustering Cryptocurrency Market Regimes Using Data Mining Techniques:** Built an end-to-end data mining pipeline on Bitcoin historical data including feature engineering, preprocessing, and K-Means clustering to identify market regimes such as bullish, bearish, and volatile phases.
- ❖ **Rear Obstacle Distance Detection:** Built a real-time computer vision app using YOLOv8 and focal length-based distance estimation to detect rear obstacles with color-coded proximity alerts. Developed a Flask web server with live video streaming and mobile-responsive GUI using multi-threading for simultaneous frame processing.
- ❖ **Poetry Application:** Desktop app in WPF (C#) for creating, managing, and displaying poetry, storing data in SQL Server with user-friendly interface.

Experience:

Remote Intern — Nepta Solutions (UK)

Jun 2025 – Aug 2025

Worked on integrating Sage 50 Accounts with Woo-Commerce using C#, developing background services for automated product and customer sync. Gained hands-on experience with Sage SDO, APIs, data mapping, and real-world e-commerce integration workflows.